

# **Cloud Computing Course**

## **Assignment 3 – Microsoft Azure Fundamentals & Practical**

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### **Submission Instructions**

1. Create a single PDF document containing all answers and screenshots.
2. Mention question numbers clearly.
3. Attach screenshots for every practical task.
4. Ensure screenshots clearly show successful completion of the task.
5. Total Marks: 50 + 10 Bonus Marks.

### **Section A – Azure Resource Group (10 Marks)**

- Q1. Create a Resource Group named azure-lab-rg (or similar). Attach screenshots showing successful creation. (5 Marks)
- Q2. Explain the purpose of a Resource Group in Azure in 3–4 lines. (5 Marks)

### **Section B – Azure Virtual Machine (15 Marks)**

- Q3. Create an Ubuntu Virtual Machine inside the Resource Group. Attach screenshots of VM creation and deployment. (5 Marks)
- Q4. Record and explain: VM Name, Region, VM Size, Public IP Address and Operating System. (5 Marks)
- Q5. Connect to the VM using SSH and attach screenshots showing successful login. (5 Marks)

### **Section C – Blob Storage (10 Marks)**

- Q6. Create a Storage Account and attach screenshots. (3 Marks)
- Q7. Create a Blob Container inside the Storage Account and attach screenshots. (3 Marks)
- Q8. Upload at least three files into the Blob Container and attach screenshots showing successful upload. (4 Marks)

### **Section D – NSG & Networking (10 Marks)**

- Q9. Create an NSG rule allowing TCP Port 3306 (MySQL). Attach screenshots of the rule configuration and creation. (5 Marks)
- Q10. Explain the purpose of an NSG and why Port 3306 might need to be opened. (5 Marks)

### **Section E – Azure Monitor Alerts (5 Marks)**

- Q11. Create an Azure Monitor Alert when CPU Utilization exceeds 60%. Attach screenshots of the alert rule. (2 Marks)
- Q12. Create an Azure Monitor Alert when available memory falls below 1 GB. Attach screenshots of the alert rule. (3 Marks)

### **Bonus Challenge (10 Marks)**

Design and draw an Azure architecture containing Resource Group, Azure VM, NSG, Blob Storage and Azure Monitor.

Show how all services interact and explain the role of each component.